



# DevOps Foundation

***“In the new world, it is not the big fish which eats the small fish, it’s the fast fish which eats the slow fish.” - Klaus Schwab (Founder of World Economic Forum)***



# Agile -Commitment

“Our highest priority is to satisfy the customer through early and continuous delivery of valuable software”

“Working Software in every short release cycle”

“Delivery Feedback, Fail early”



Waterfall



Agile



DevOps



SecDevOps

# Product World

“Product” is a Living Thing.... It Evolves....

- Grows in Size..
- Number of Users and Usage
- Transactions
- Logs Size
- DB size
- Interfaces
- Functionality & Features
- New Business Requirements/Project Features
- Technology Enhancements & Updates
- Underline Technologies such as OS, .IOS versions etc..
- Non Functional
- Performance and Scalability
- Security Updates
- Systems Update / Infrastructure Changes
- High Availability/Stability
- Infrastructure governance
- Resource Management

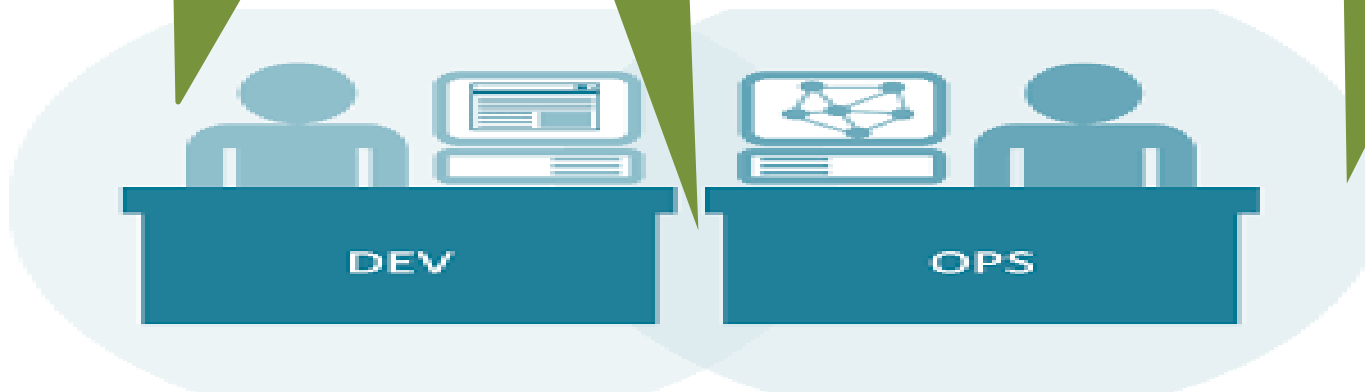


# Product Lifecycle

- Functionality & Features
- New Business Requirements/Project Features
- Technology Enhancements & Updates
- Interfaces UPDATES

- Performance and Scalability
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# Current Challenge

From Developer to Operations:

As a developer I have always dabbled lightly in operations. I always wanted to focus on making my code great and let an operations team worry about setting up the production infrastructure.

As a Operation Engineer I always worry about System stability, never reach out to developer to communicate production scenarios and help him consider operation issues while development.





# Technology Organizations Create Impenetrable Silos

"Our growth is slowing... Why can't we get new products out the door faster than our competitors?"

"Our headcount has been flat, and our engineers have a backlog of work."

"We keep slipping dates because we can't deliver what we promised."

"Clients are angry. We're spending 80% of time keeping systems up. Don't throw stuff over the fence and run. My team is ready to quit!"

DevOps

Source: CiterOpenDays OC 2015 word cloud

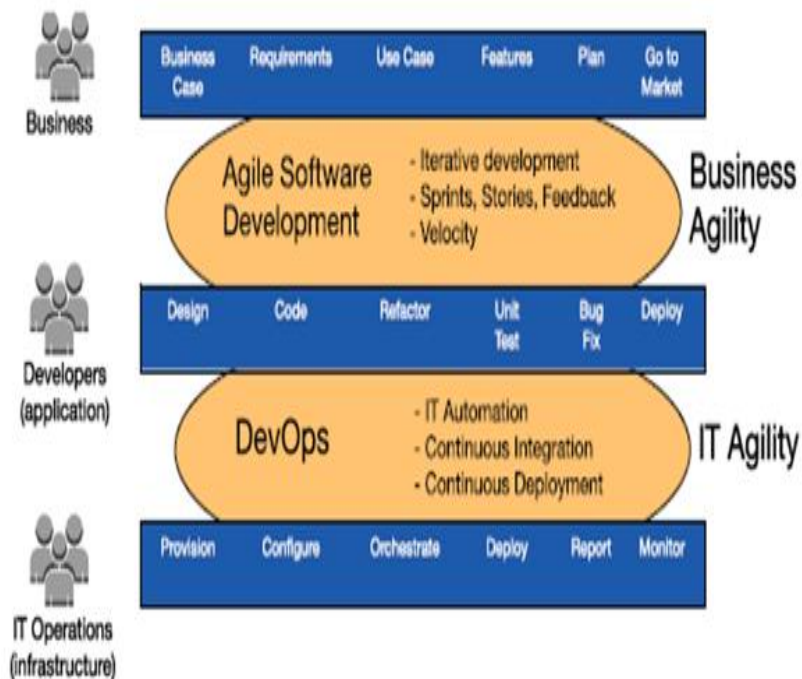
- Source: DevOps Days DC 2018 word cloud from Open Space.

**DevOps is a business-driven approach to rapidly deliver solutions through collaboration.**



# What is DevOps ?(cont'd)

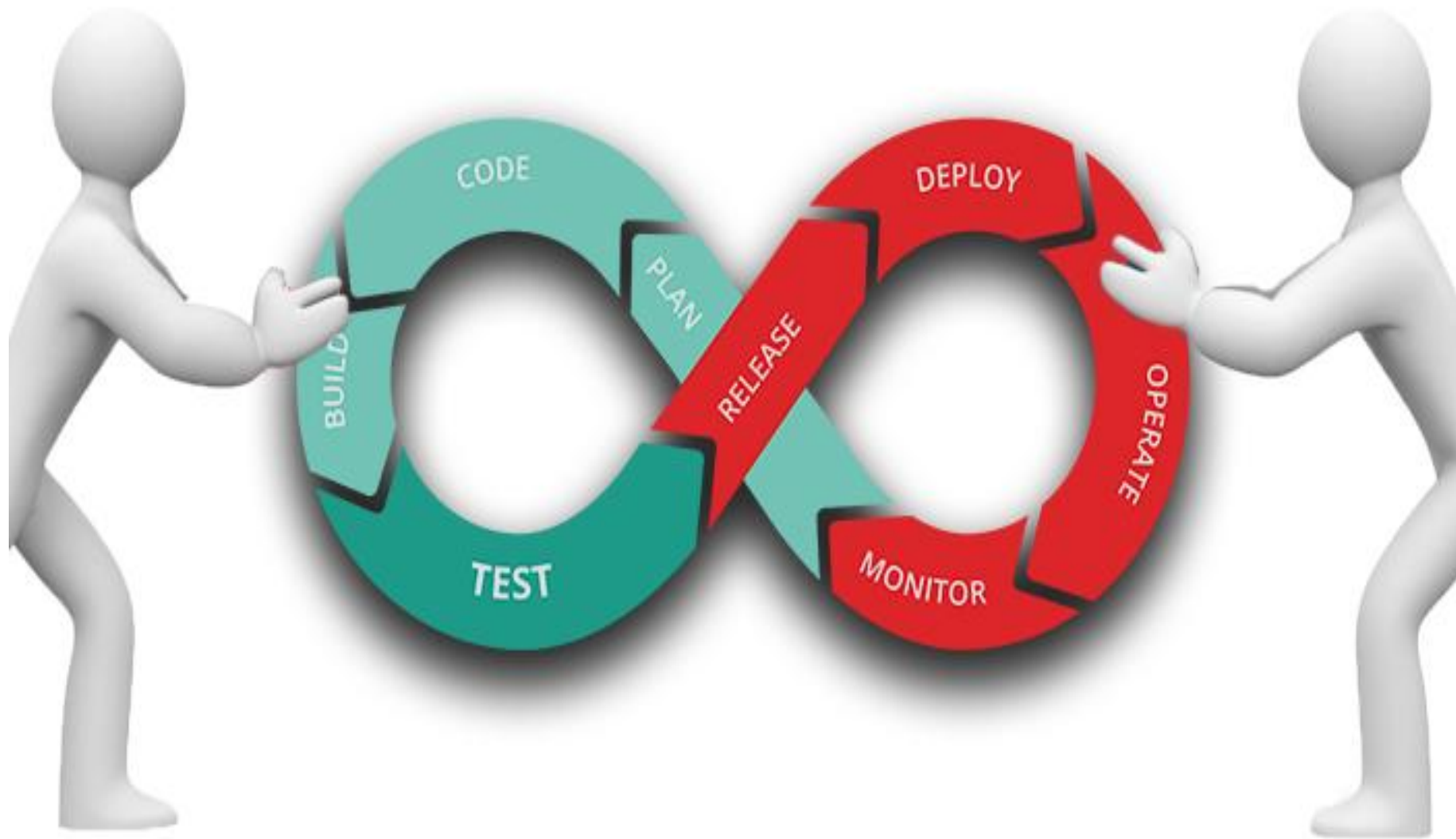
It involves getting key technology stakeholder working together more effectively and combining this with best in class release and infrastructure automation in order to speed up delivery and focus everyone on the real business goals of software.



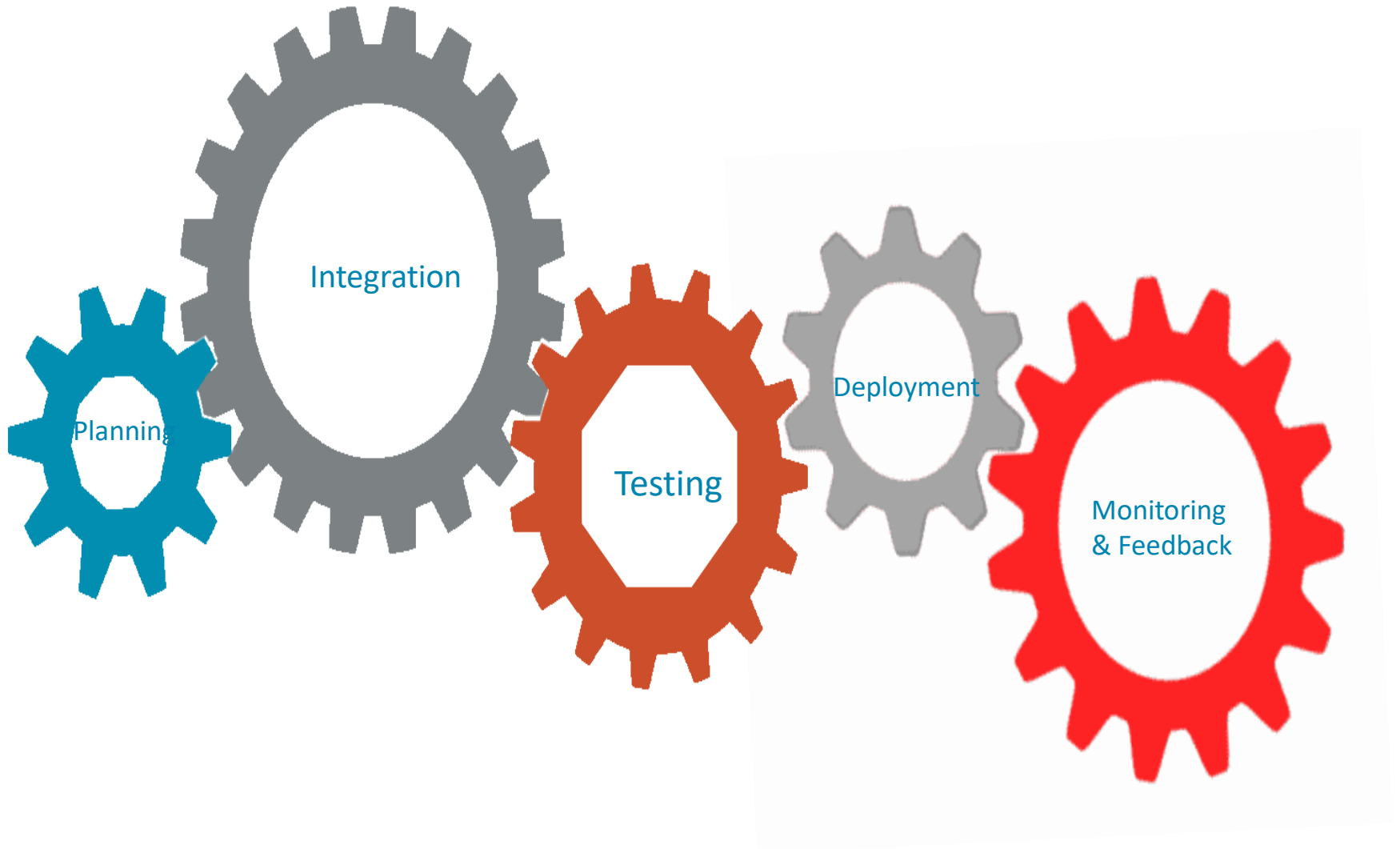
When Devops is adopted successfully, the organization is empowered to deliver software better- giving them an outsized ability to beat competition get to market faster and innovate much more effectively.

# DEVELOPMENT

# OPERATIONS



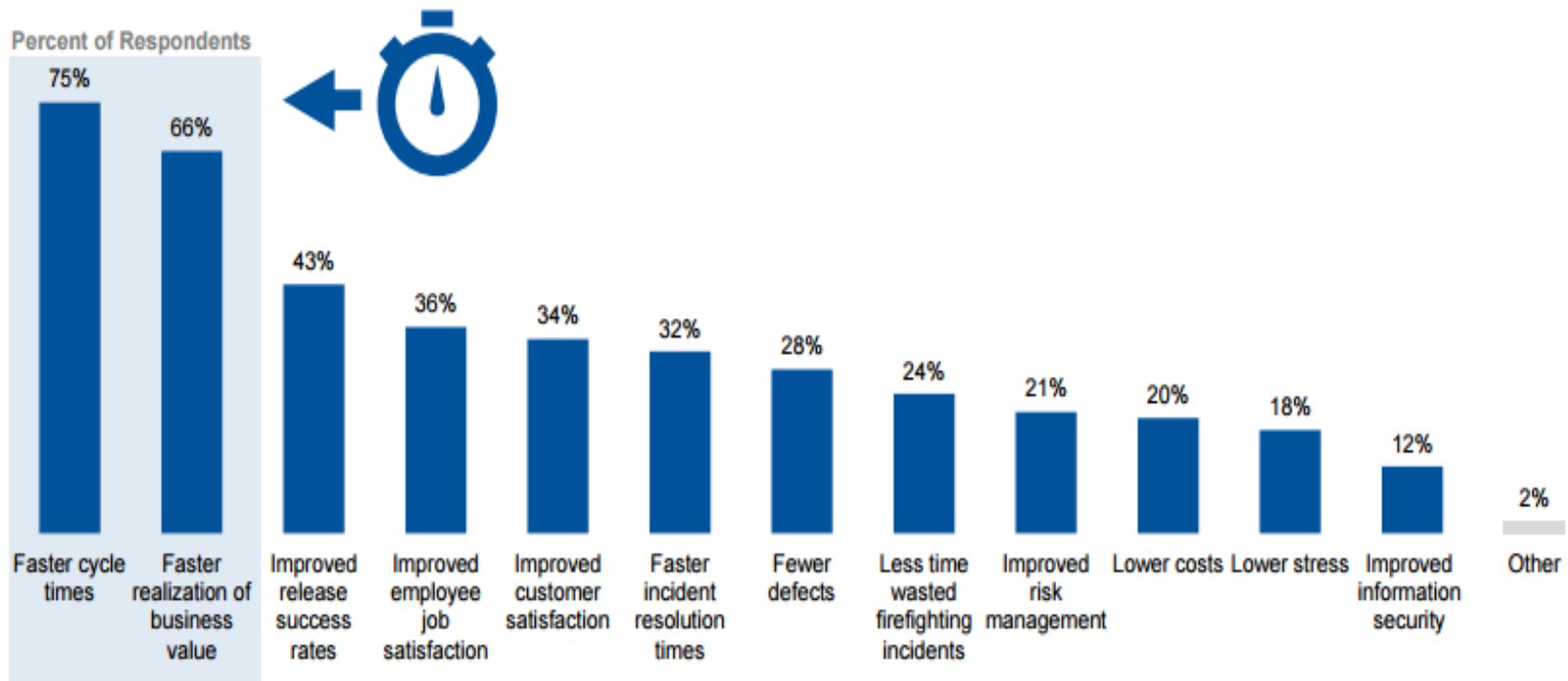
# 5Cs of Devops (Cont..)



# The Value Proposition

## Faster Cycles Times and Realization of Business Value Are the Most Frequent Organizational Outcomes From DevOps

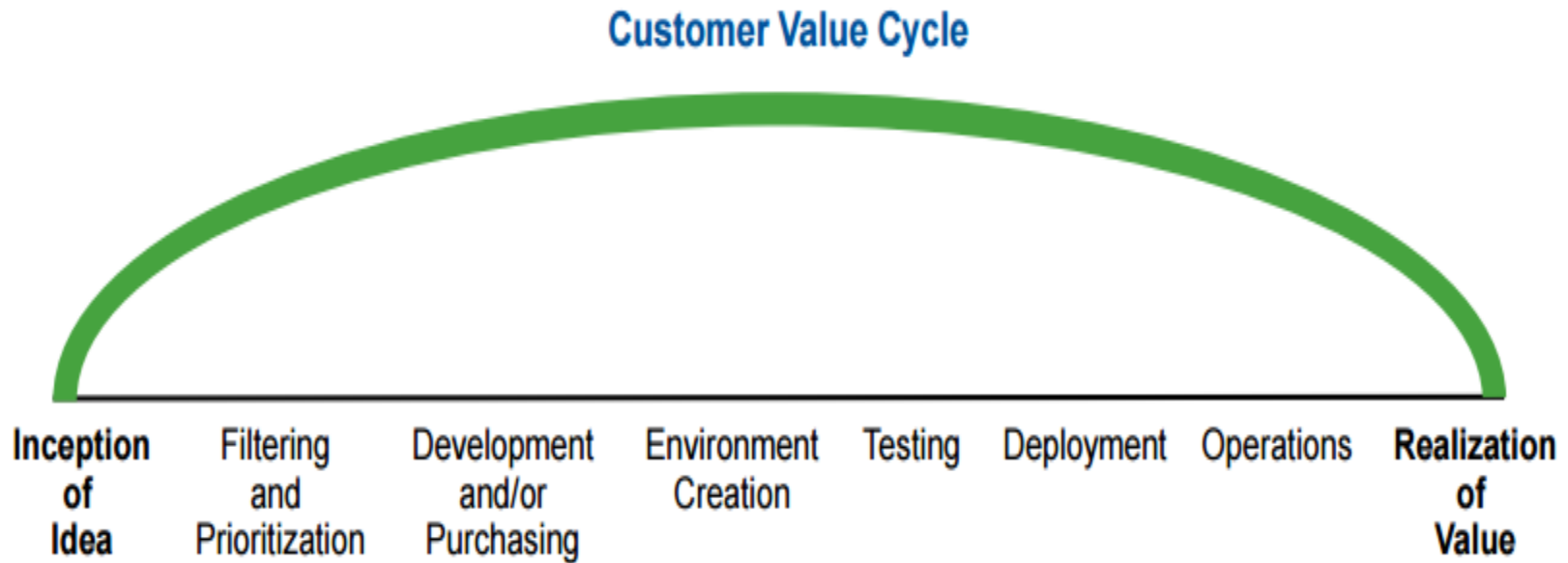
Outcomes From DevOps:



Base: n = 95 Gartner Research Circle Members who use DevOps approach  
Q06. Which of these outcomes has DevOps created for your organization?

## The Value Proposition (cont'd)

### Improve Mean Time To Value (MTTV)



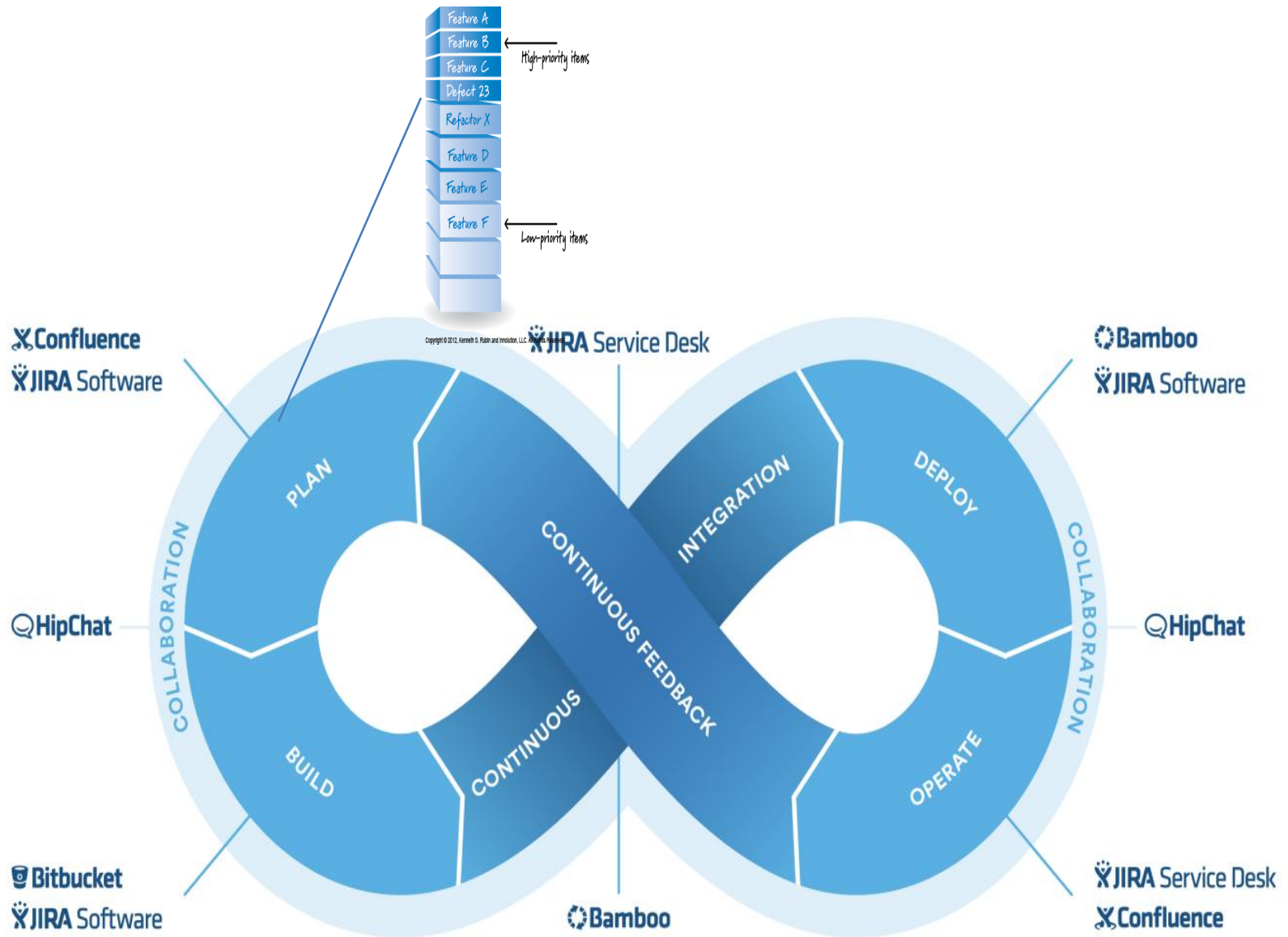


# 5 C's of DevOps

- Continuous Planning
- Continuous Integration
- Continuous Testing
- Continuous Deployment
- Continuous Feedback & Monitoring

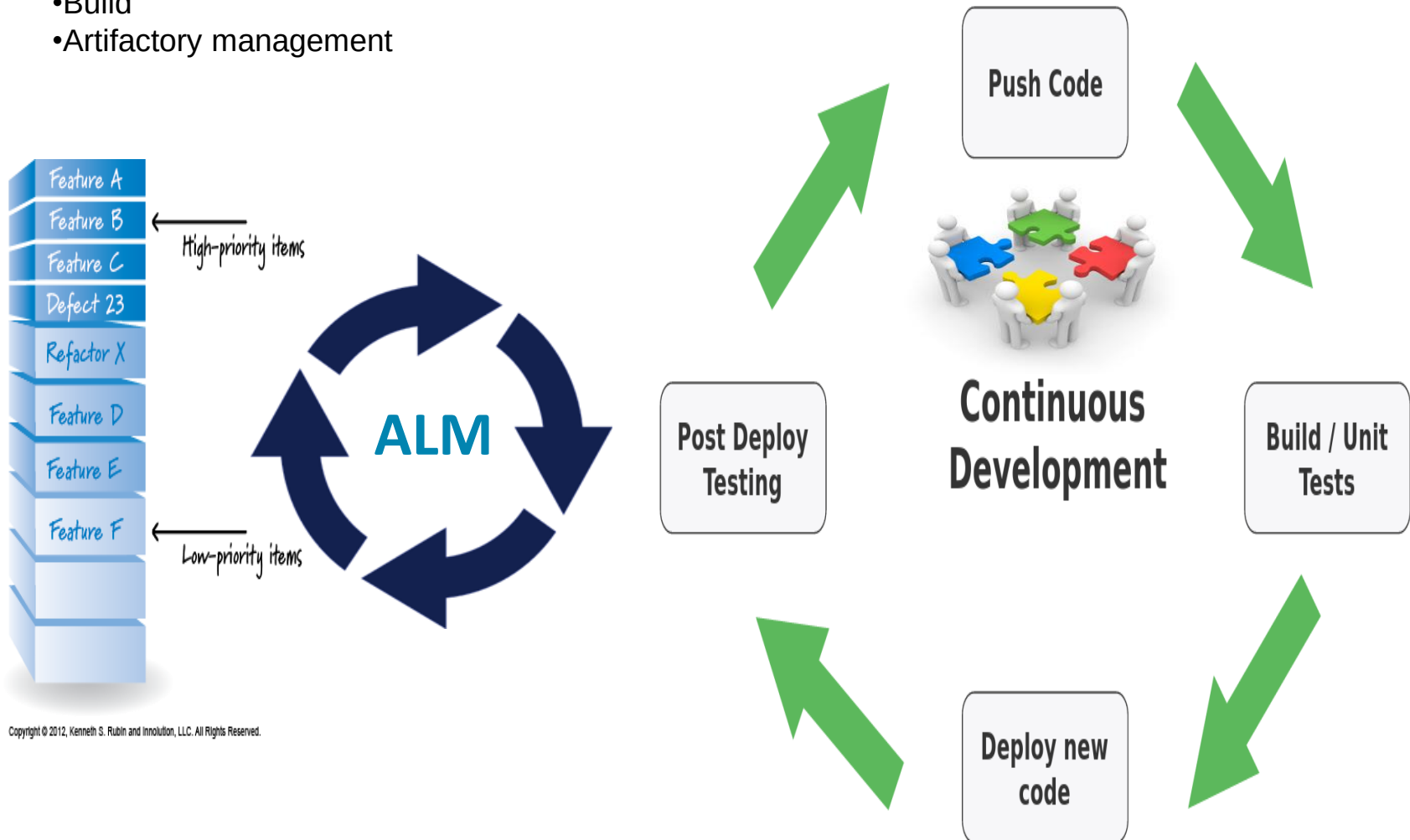


# Continuous Planning

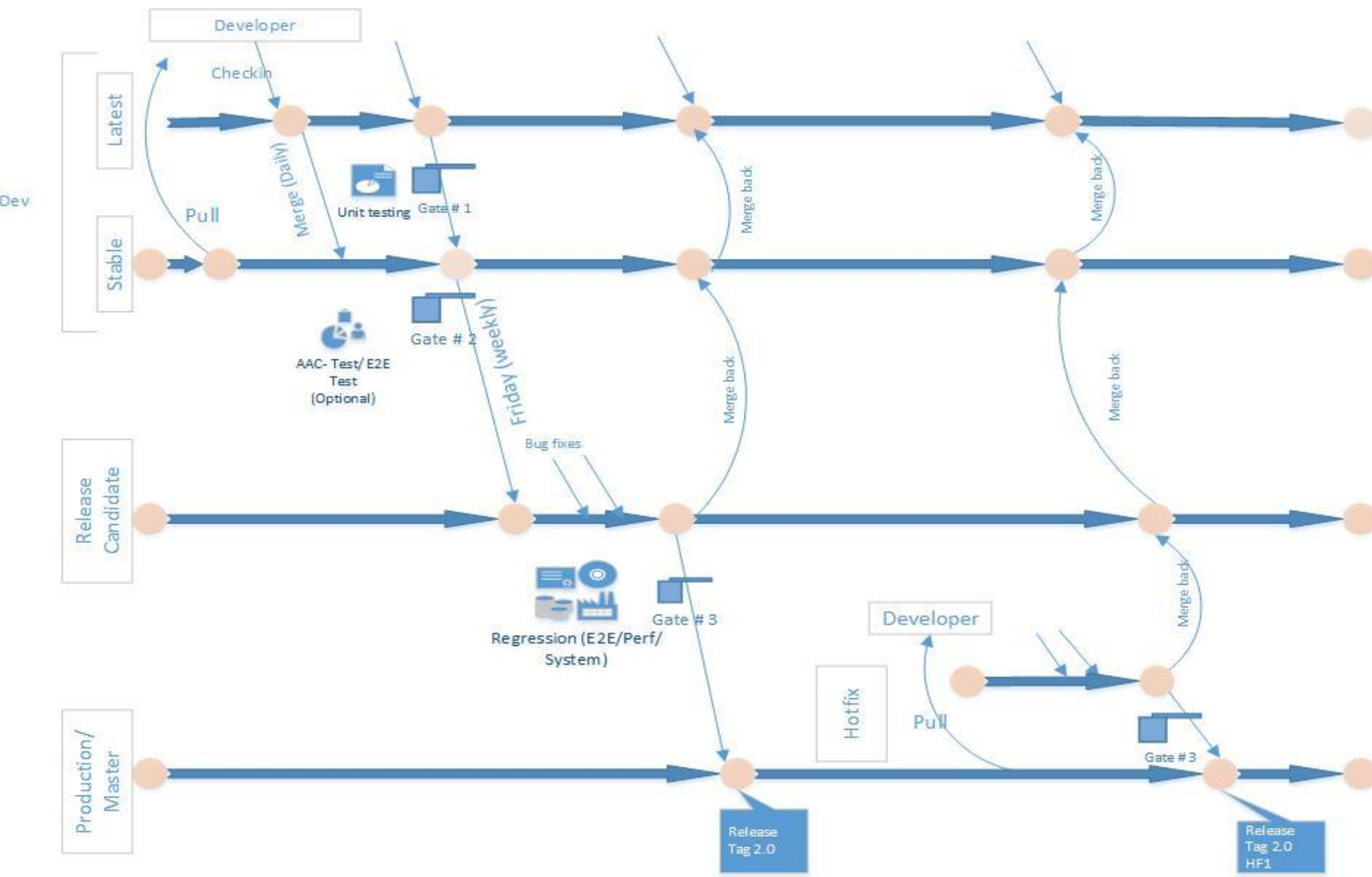


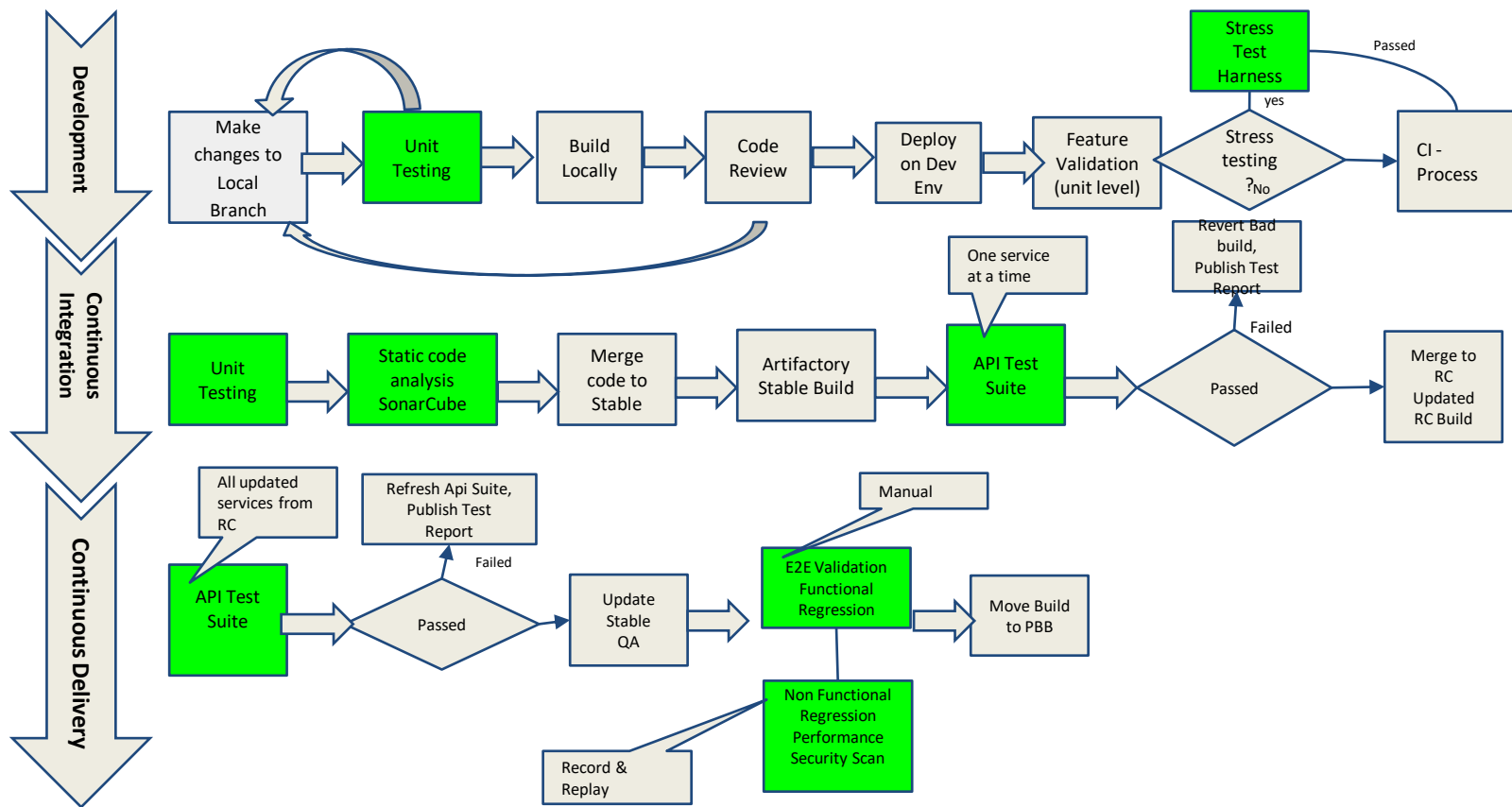
# Continuous Integration (Development)

- Version controlling tool chain Git hub etc.
- Release Management Process
- Code review mechanism
- Static code analysis using Sonar cube
- Build
- Artifactory management

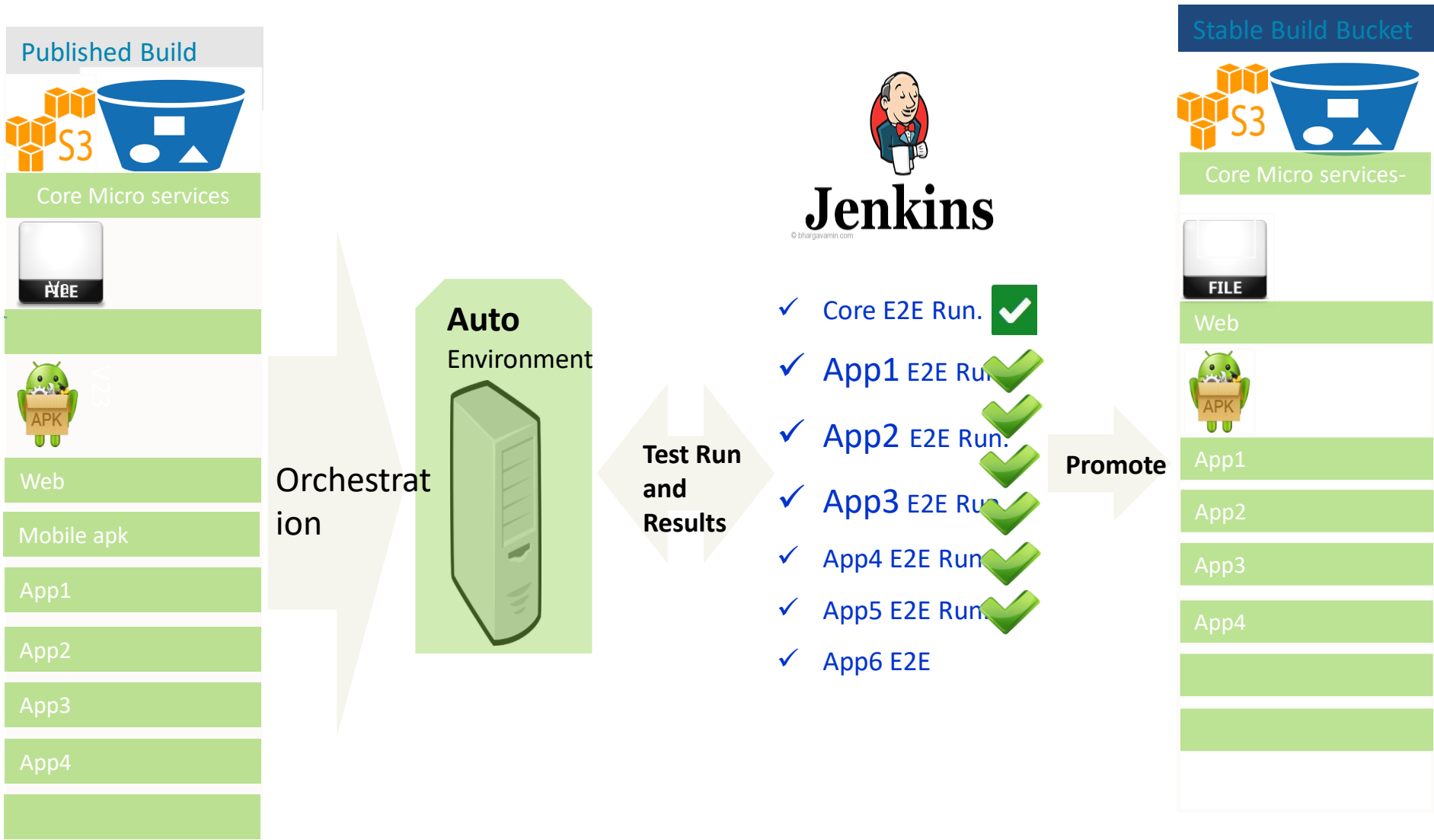


# Release Management- Gitflow





# Gate # 2. Continuous Testing





# Continuous Deployment

- Blue- Green Deployment
- Rolling out Deployment
- Canary Deployment
- Feature Toggle
- Multi Region Deployment

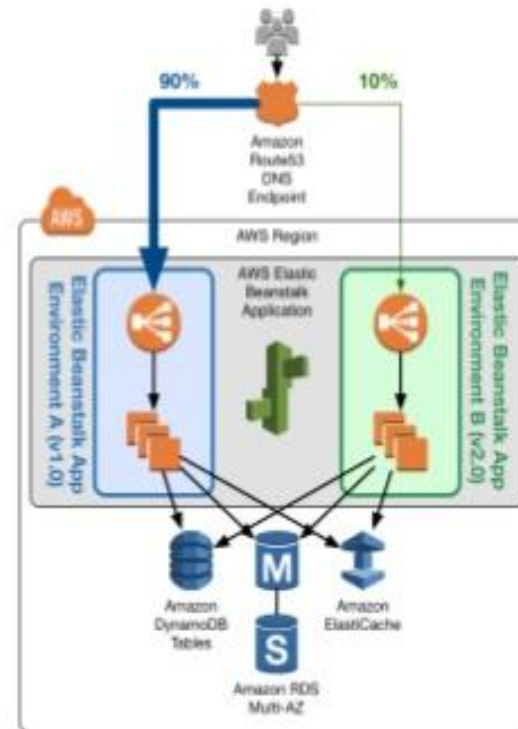
# Blue Green Deployment

- Case study: Umair will show a auto Deployment utility and how it works

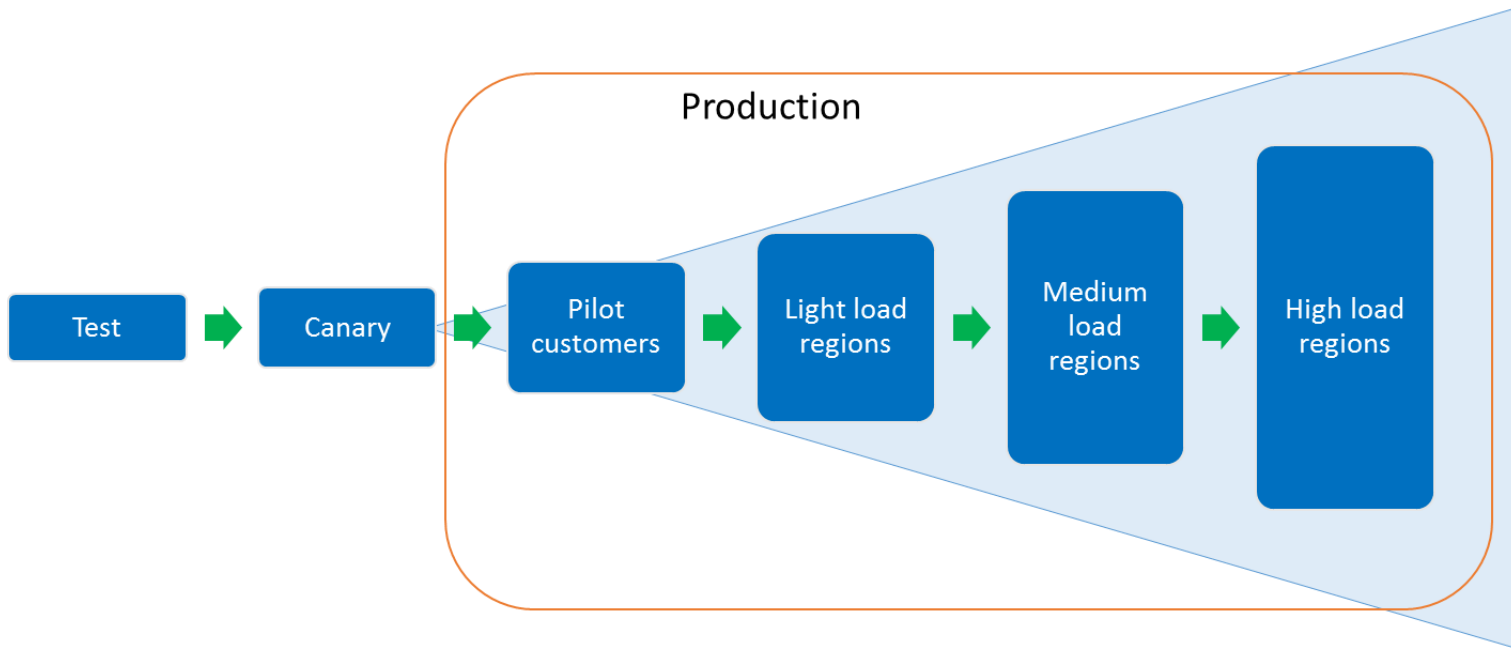
## Patterns: classic DNS cutover

### Deployment process:

- Start with **current** app environment
- Deploy a **new** app environment
- **Test** the green stack
- Gradually **cut traffic over** via DNS
- **Monitor** your environments
- If needed, **roll back** to blue



# Rolling out Deployment



# Carnary Release(Deployment)

- <https://code.fb.com/web/rapid-release-at-massive-scale/>
- <https://www.infoq.com/news/2017/09/facebook-release-scale>
- <https://instagram-engineering.com/continuous-deployment-at-instagram-1e18548f01d1>

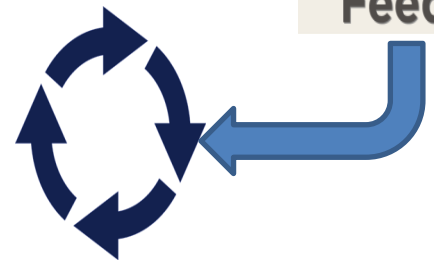
# Continuous Monitoring & Feedback



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# 5 Cs Implementation-

Quality Gate #1

Quality Gate #2

Quality Gate #3

Quality Gate #4

CAC

AAC

RAC

PAC

**Continuous Planning**  
(Pre Req)

**Continuous Development**  
**Integration (Daily)**

**Continuous Testing**  
Weekly/Sprint/  
Rel. Regression

**Continuous Deployment & Monitoring**



PO-Business

APP1

CORE

App2

App3

Cloud

Web



- Build Process
- Installer/Upgrader
- Local QA Environment.
- AAT/Sanity Run
- Manual Validation
- Scope Testing
- Automation Test Run (Component)



- QA Env.
- Core E2E Run.
- App1 E2E Run
- App2 E2E Run.
- App3 E2E Run.
- Cloud E2E Run.
- Web E2E Run.



- Staging Env.
- Production Simulated Env, Integrated with all external dependencies, Validate Deployment - SIT -E2E

- Performance Box
- Performance Testing
- Security Testing



devops



PO-Tech



Architect